Neil Roy

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Education

University of California, Santa Barbara — Santa Barbara, CA

M.S. in Computer Science (5-Year B.S./M.S. Program) *GPA: 3.87* — *Sept. 2025 – June 2027 (Expected)*
Grad Coursework: Cryptography, Security & Threat Detection, Agentic AI, Advanced Computer Architecture

B.S. in Computer Science, ***Distinction in the Major*** *GPA: 3.77* — *Sept. 2022 – June 2026*
Coursework: Operating Systems, Distributed Systems, Data Structures & Algorithms, Databases

Experience

Qualcomm — June 2026 – Sept. 2026

Modem Software Engineering Intern — Layer 2 Firmware Downlink (L2 FW DL) — *San Diego, CA*

- Built a **Python** crash/core-dump analyzer for **5G NR** and **LTE** modem firmware, run on **25+ crashes daily** across **5 product lines** and used by the entire **15-engineer** team
- Automated root-cause triage using **ELF** symbols and **Trace32** to reconstruct call stacks and L2 state, decoding transport blocks into **MAC**, **RLC**, and **PDCP** headers
- Replaced manual Trace32 loading/querying, saving **5–10 min per dump** (**1–2 eng-hours/day**), easing sharing
- Designed a **modular, extensible architecture** with a centralized config and per-product-line classes, so new product lines onboard with minimal code and other modem teams can reuse the framework as a baseline
- Integrated the analyzer into **agentic LLM workflows** (**Claude**, **Codex**, QGenie via API) and internal automation, structuring output to **minimize token usage** for automated crash summaries
- Surfaced naming inconsistencies and unintended **cross-product-line differences** in the firmware codebase
- Debugged crashes across time zones; presented to the **Modem Software org** and a **company-wide poster session**

Mysten Labs (Sui / Walrus) — Sept. 2025 – June 2026

Software Engineering Intern — Web3 Storage Capstone — *Santa Barbara, CA*

- Led **5 engineers** to build and ship Infinity Storage, a **hybrid decentralized storage platform** on **Walrus**/**Sui**
- Owned the **TypeScript/Next.js** backend: **Walrus** integration, **REST APIs**, and **Stripe** payments
- Worked with Walrus co-author **Alberto Sonnino** on architecture, APIs, storage efficiency, and **end-to-end encryption**
- Managed the project on **GitHub**: scoped issues, led **code reviews**, and assigned client/server-side tasks

Research

Fftech Research Lab, UC Santa Barbara (PI: Prof. Dahlia Malkhi) — Mar. 2026 – Present

Graduate Student Researcher — Decentralized Storage & Distributed Systems — *Santa Barbara, CA*

- Designing and implementing a new **information dispersal algorithm** for **Walrus** with Rafture authors Rithwik Kerur and Dahlia Malkhi, made **Byzantine fault-tolerant**, targeting **2× lower storage cost**
- Replacing Walrus's 2D Reed–Solomon (RedStuff) encoding with a single static **MDS erasure code** ($k = 2F{+}1$ of $m = kN$ fragments), **Merkle-proof** fragment verification, and adaptive per-shard dissemination
- Developing **post-dissemination pruning** driven by **quorum certificates**, converging to one fragment per shard, and integrating it across Walrus's **Rust** codebase (encoding, SDK, storage nodes, Sui Move contracts)

Projects

User-Level Thread Library | *C++, POSIX Signals, Linux* — Sept. 2025 – Dec. 2025

- Implemented a **user-level threading library** from scratch using `sigsetjmp`, stack allocation, and context setup
- Built a **preemptive round-robin scheduler** driven by `SIGALRM` timer interrupts via `setitimer` for time slicing
- Designed **semaphores** with wait queues and signal-masked critical sections to prevent **race conditions**

Scrypt Interpreter | *C++, Git, Docker* — Sept. 2023 – Dec. 2023

- Built a **Turing-complete** interpreter in **C++** (team of 4) with a **lexer**, **parser**, and **AST** evaluator

Technical Skills

Languages: Python, C, C++, Rust, TypeScript, JavaScript, Java, SQL
Systems & Embedded: Embedded Firmware, Crash-Dump Analysis, Trace32, ELF, Linux, Multithreading
Wireless & Networking: 5G NR, LTE, 3GPP Layer 2 (MAC, RLC, PDCP), Cellular Modem Software
Distributed Systems: Erasure Coding (Reed–Solomon), Byzantine Fault Tolerance, Merkle Trees, Sui, Walrus
AI & Automation: Agentic AI Workflows, LLM APIs (Claude, Codex), Token Optimization, Prompt Engineering
Web & Tools: React, Next.js, Node.js, REST APIs, Stripe, AWS (S3), GCP, Docker, Git, GitHub Actions, Perforce